December 2017 exam

Solutions

Q1 (a) $p_{53} = 0.9591$

$q_{53} = 1 - p_{53} = \underline{0.0409}$

(b) $p_{56} = 0.9264$

$p_{57} = 0.9151$

$p_{58} = 0.9036$

${}_3p_{50} = p_{56}\ p_{57}\ p_{58} = \underline{0.766026}$

(c) $v\, q_{55} + v^2\, p_{55}\, q_{56} = 0.115415$ at 9%

so $\underline{\$115,415}$

(d) Repeat with $i_2 = 18.81\%$

${}^2A = 0.101486$

Then ${}^2A - (A)^2 = 0.088166 = \text{variance}$

$SD = \sqrt{\ \ }$

$= 0.296927$

so $\underline{\$296,927}$

Q2

Claim expenses $= 0.2\% \times 100,000$

$= £200$

So use £100,200 for claim

Value of benefits $= 100,200 \left(v q_{48} + v^2 {}_1p_{48} q_{49} \right)$

$= £2,415.82$

Value of premiums $= 300 \left(1 + v p_{48} \right) \times 0.98$

$= 294 \times 1.9686$

$= £578.78$

Reserve $= 2415.82 - 578.78$

$= £1837$

Q3

(a) $\mu_x = \lim_{h \to 0} \left(\frac{1}{h} \Pr [0 < T_x \leq h] \right)$

where T_x is the time until death
(as a real number)

(b) $S_0(5) = 7/8$

$S_0(39) = 5/8$

$P = 7/8 - 5/8$

$= \frac{1}{4}$

Q4 (a) At start,

Value of benefits $= 200{,}000\, v^{20}\, \frac{L_{65}}{L_{[45]}}$ (8821.261 / 9798.084)

$= £82{,}177.47$

$\ddot{a}_{[45]:\overline{20}|} = 13.785$

$\therefore$ Premium $= 82{,}177 / 13.785$

$= £5{,}961.37$

(b) Reserve per policy at age 56

$= 200{,}000\, v^{9}\, \frac{L_{65}}{L_{56}} - P\, \ddot{a}_{56:\overline{9}|}$ (L_{56}: 5915.104; $\ddot{a}_{56:\overline{9}|}$: 7.542)

$= £85{,}310.16$

Reserve per policy at age 57

$= 200{,}000\, v^{8}\, \frac{L_{65}}{L_{57}} - P\, \ddot{a}_{57:\overline{8}|}$ (L_{57}: 9467.2901; $\ddot{a}_{57:\overline{8}|}$: 6.838)

$= £95{,}402.03$

Q4(b) continued

At year end, have

$$17000\,(85{,}310.16 + 5961.37) \times 1.04$$

$$= £1{,}613{,}680{,}612 \quad (1)$$

At year end, ~~need~~ Lives $= 17000 - 103$

$$= 16{,}897$$

Need $16{,}897 \times a_{5,402}$

$$= £1{,}612{,}008{,}072 \quad (2)$$

Profit $= (1) - (2) = £1{,}672{,}540$

(c) $E(\text{deaths}) = 17000\, {}_{0.005025}q_{56}$

$$= 85.425$$

< actual deaths

For a pure endowment, there is a profit on deaths.

Q5

x	l_x
100	1501
101	1013
102	663
103	423

(a) UDD: $l_{100.1} = 0.1 \times 1013 + 0.9 \times 1501$

$= 1452.2$

$l_{102.6} = 519$

$q = 1 - \frac{519}{1452.2}$

$q = 0.6426$

(b) Constant μ: $l_{100.1} = l_{100} (p_{100})^{0.1}$

$= 1443.124$

$l_{102.6} = l_{102} (p_{100})^{0.6}$

$= 506.302$

$q = 0.649162$

(c) $p_{100} = 0.674883$

$= e^{-\mu}$

$\therefore \mu = -\log p = 0.3932$

(d) Between μ_{100} and μ_{101} in tables.

Q6

(a) $(\bar{I}\bar{A})_{35:\overline{20}|}$

(b) Value of term part

$$= \int_0^{20} t\, v^t\, {}_tp_x\, \mu_{x+t}\, dt \qquad (1)$$

$${}_tp_x = e^{-\mu t}$$

$$\therefore (1) = \int_0^{20} t\, e^{-\delta t} e^{-\mu t}\, 0.0008\, dt$$

$$= 0.0008 \int_0^{20} t\, e^{-(\mu+\delta)t}\, dt$$

Integral [by parts] $= -\left[\frac{t\, e^{-(\mu+\delta)t}}{(\mu+\delta)}\right]_0^{20} + \int_0^{20} \frac{e^{-(\mu+\delta)t}}{(\mu+\delta)}\, dt$

$$= \frac{-20\, e^{-20(\mu+\delta)}}{(\mu+\delta)} - \left[\frac{e^{-(\mu+\delta)t}}{(\mu+\delta)^2}\right]_0^{20}$$

$$= \frac{1}{(\mu+\delta)^2} - \frac{e^{-20(\mu+\delta)}}{\mu+\delta}\left[20 + \frac{1}{\mu+\delta}\right]$$

$$= 173.4043$$

$$\therefore (1) = 173.4043 \times 0.0008$$

$$= \underline{0.138723}$$

Q6 (continued)

Pure Endowment part $= 20 e^{-20(\delta)} \; e^{-\mu t}$

$= 20 e^{-(\mu+\delta)20}$

$= 16.11471$

$(\bar{I}\bar{A})_{35:20} = 16.11471 + 0.138723$

$= \underline{\underline{16.253}}$

(c) If mortality were zero,

Answer $= 20 e^{-20\delta}$

$= \underline{\underline{16.375}}$

Actual answer is slightly less because on death, benefit < 20.

Q7

(a) Benefit at 65 = 200,000 (1 + 60%)

= £320,000

Value of benefit of 200,100 (including death expenses)

$= 200{,}100\ A^{1}_{[40]:\overline{25}|}$ (0.38846)

= £77,831

Value of extra benefit at 65 = (320,000 − 200,100) $\frac{D_{65}}{D_{[40]}}$ (D_{65} = 689.23, $D_{[40]}$ = 2052.54)

= £40,262

Value of fixed expenses = $600 + 80\, a^{12.930}_{[40]:\overline{20}|}$

↳ $= \ddot{a}_{[40]:\overline{20}|} - 1$

= £1,634

Now $\ddot{a}^{(12)}_{[40]:\overline{20}|} = 13.930 - \frac{11}{24}\left(1 - \frac{D_{60}}{D_{[40]}}\right)$ (D_{60} = 882.85, $D_{[40]}$ = 2052.54)

= 13.669

Value of premiums per year = ~~P (13.669 − 0.8~~

$= P(0.988 \times 13.669 - 0.8)$

= 12.595 P

$\therefore$ 12.595 P = 77,831 + 40,262 + 1,634

P = £9506 per year

= £792.13 per month

Q7 (b) Not a problem – death benefit need not equal survival benefit.

People with death benefit will usually have paid fewer premiums.